2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.203 Computer Networks

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Compare and contrast connectionless and connection-oriented service mode by

i. Explaining the purpose and main features of both **[2 marks]**
ii. Providing a description of the communication phases of both modes **[2 marks]**
iii. Giving examples of protocols for either mode **[1 marks]**

You are asked to design a communication service for delivering video streams over the Internet. Will your service use a connectionless or connection-oriented mode? Explain your answer **[2 marks]**

1.b Since the costs of cables are high, multiplexing is being used to make optimal use of the resources over the physical medium. With multiplexing multiple conversations are transmitted over one physical medium. Describe and contrast Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM). In the case of TDM there is a synchronous and an asynchronous mode. Briefly explain how they work and outline their advantages and disadvantages.

[5 Marks]

1.c Two-dimensional parity schemes are used for the detection and correction of errors. In this scheme the Sender applies an algorithm that generates redundant bits, the receiver applies the same algorithm to the message, compares result with extra bits received to decide if the message is correct or was corrupted in the transmission.
Suppose, a two dimensional odd parity scheme is used. The two- dimensional data bits of a packet are shown below:

```
0011
0101
1011
1101
```

i. Calculate the parity bits for each row and each column assuming odd parity is being used **[2 marks]**
ii. Show an example where there is a bit error that can be detected and corrected. Explain your answer **[2 marks]**
iii. Show an example where errors can be detected but not corrected. Explain why this is so **[2 marks]**

Question 1 continues on next page...

Question 1 continued…

1.d You are asked by your employer to set up an office network based on 802.3 Ethernet. They are considering a standard Ethernet or Switched-Ethernet set-up.

i. Outline the Switched-Ethernet set-up of with respect to connecting hosts to the network infrastructure (use a diagram if you wish). What role does CSMA/CD play in this context? **[2 marks]**

ii. One advantage of Switched-Ethernet is that switches are self-learning. Describe what this means. How does a switch deal with frames with an unknown destination address? **[2 marks]**

iii. The switches you are contemplating have got 10 ports. You want to connect 15 hosts. What set-up do you suggest (use a diagram to illustrate it)? Describe the structure and forwarding of frames in your design and discuss advantages and disadvantages. **[3 marks]**

Total 25 marks

Question 2

2.a IPv4 addresses were initially *class based* but are now said to be *classless*.

i. Being careful to explain the difference, describe what we mean by *class based* and *classless* IPv4 addresses

[4 marks]

ii. Outline why the change from *class based* to *classless* addresses was made

[2 marks]

iii. What is the purpose of a subnet mask and why is one required when using classless addressing? Provide an example to show how a subnet mask is used.

[2 marks]

2.b Identify the type or significance of each of the following IPv4 addresses and outline where a packet sent to each of these addresses would be delivered.

i. 224.0.0.1

ii. 0.0.255.255

iii. 192.168.5.255

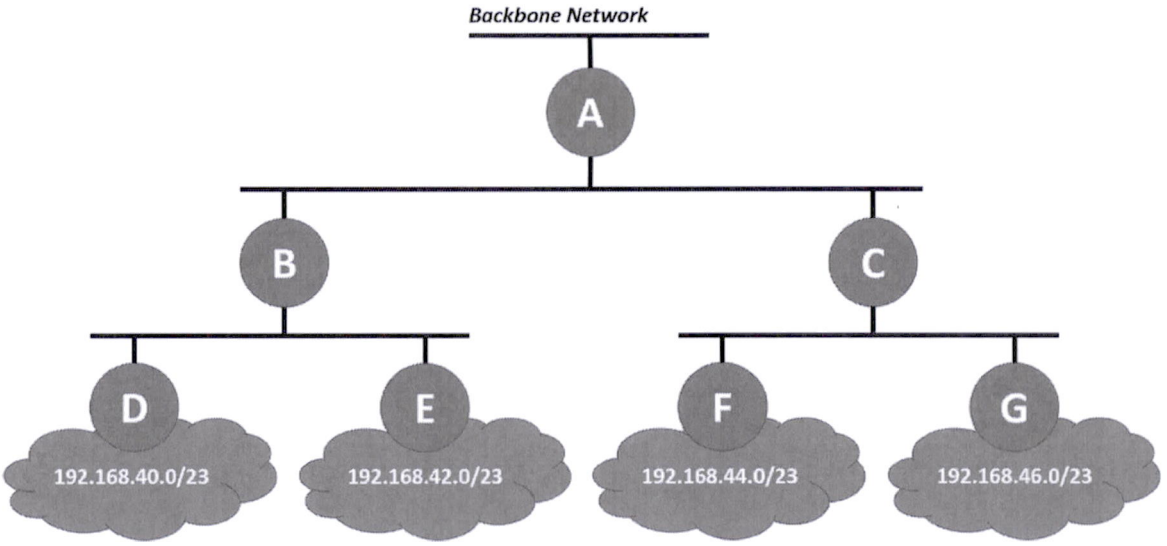
iv. 255.255.255.255

[4 marks]

Question 2 continues next page...

Question 2 continued...

2.c Given the following network structure:



i. Considering the network behind router F, 192.168.42.0/23, and explaining your answer, determine how many host addresses are available on this network. Your answer should include the host address assigned to the router itself within the total given.

[2 marks]

ii. Assuming routers A to G are configured to use variable length subnet masks (VLSM), classless interdomain routing (CIDR), and route summarisation (or prefix aggregation), state the prefix announcements each of the routers A, B and C pass upstream toward the backbone network.

[4 marks]

iii. During normal operation, what prefixes and associated next hop routers would be held in the forwarding table of router B?

[4 marks]

iv. Prefix aggregation allows the possibility of a packet arriving at a router destined for a non-existent network. Outline why this is the case, what action a router should take if this happens, and why this action is mandated or demanded.

[3 marks]

Total 25 marks

Question 3

3.a TCP uses a timer to detect the non-delivery of segments. What does TCP normally assume the loss of segments to be an indication of and what is its response?

[2 marks]

3.b The actions of TCP senders are governed by the acknowledgements returned by receivers

i. What are *duplicate ACKs* and *cumulative ACKs* and how are they used in a TCP implementation supporting *fast retransmission*?

[4 marks]

ii. A modern TCP sender is attempting to send a sequence of segments with sequence numbers 10, 11, 12, 13, 14, 15. If segment 11 is lost and doesn't reach the receiver but all others are delivered what sequence of acknowledgements will be returned to the sender? **[4 marks]**

iii. Draw a Time-Sequence diagram that clearly illustrates the exchange of TCP messages in your previous answer (3.b ii)

[3 marks]

3.c TCP connections are initiated by what is known as a *three-way handshake*.

i. With the aid of an example that clearly shows the TCP messages and sequence numbers exchanged, show how this connection initiation works.

[4 marks]

ii. This three-way handshake has a resource disparity between sender and receiver that until recently left implementations open to a *SYN* or *SYN-Flood attack*. Being careful to describe how such an attack works, explain this statement and outline how current TCP implementations try to protect themselves against this type of attack.

[8 marks]

Total 25 marks

Question 4

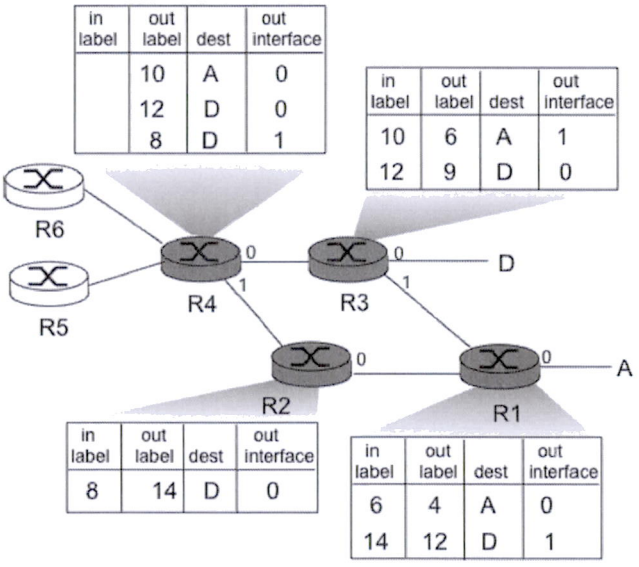
4.a Multiprotocol Label Switching (MPLS) has been proposed for high-speed IP packet forwarding. It uses fixed length identifiers rather than shortest prefix matching.

i. Does MPLS provide a connectionless or connection-oriented transmission of messages? What is the advantage of using MPLS? Explain your answer; **[2.5 marks]**

ii. The diagram below shows a router infrastructure with a set of MPLS forwarding tables. Describe the route a packet for destination A will take that arrives at R4 through R6. Use a diagram if you wish; **[2 marks]**

iii. Assume the input label of packets for destination D arriving at R4 from R6 is 4 and the output label is 12, and the input label for destination D arriving at R4 from R5 is 2, output label 8, will packets for D arriving from R4 and R6 take the same or different routes? What happens if out interface 1 becomes unavailable? Explain your answer; **[2.5 marks]**

iv. Routes to D from R4 via interface 1 are not stable. What would you suggest to ensure packets coming from R5 are forwarded to D despite the link between R4 and R2 being temporarily down? **[2 marks]**



in label	out label	dest	out interface
	10	A	0
	12	D	0
	8	D	1

in label	out label	dest	out interface
10	6	A	1
12	9	D	0

in label	out label	dest	out interface
8	14	D	0

in label	out label	dest	out interface
6	4	A	0
14	12	D	1

Question 4 continues on next page...

Question 4 continued...

4.b One of the major issues IPv6 tries to solve is the shortage of IP addresses. Describe the solution proposed by IPv6 for this issue. How is backward compatibility with IPv4 ensured? In this context also explain the changes that have been made to the IP header and what this achieves.

[6 Marks]

4.c The Domain Name System (DNS) is used to map names onto IP addresses.

i. There are two forms of queries to resolve names and find the respective IP addresses, i.e. recursive query and the iterative query. Explain how these queries work. **[3 marks]**

ii. You are a system administrator of Muncaster University and an attacker runs a Distributed Denial of Service (DOS) attack against the local DNS server. Describe what such an attack does and what effect this has on the users in your network. Also, is there a difference if recursive or iterative queries are being used in this context? **[3 marks]**

4.d The User Datagram Protocol (UDP) is an alternative to the Transmission Control Protocol (TCP) within the Internet protocol suite.

i. Briefly describe the features of UDP **[2 marks]**

ii. You are a software developer in an Internet Television service provider and are asked to write a programme that transmits short live video clips from a IP based camera to a client. Which transport protocol would you use? Explain your answer **[2 marks]**

Total 25 marks

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